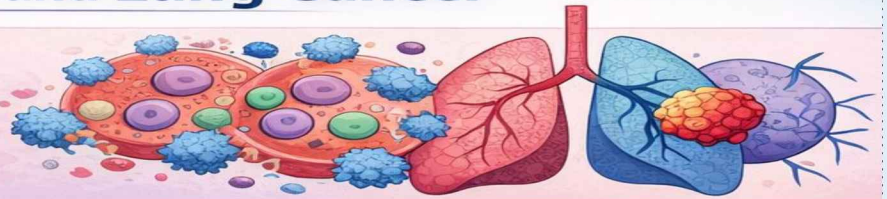
T-Cell Exhaustion in Breast and Lung Cancer

Breast Cancer & Lung Cancer

T-Cell Exhaustion

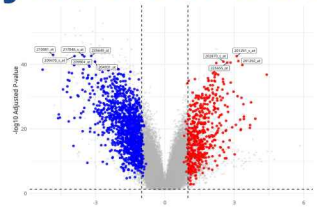
Challenges in Targeted Therapies



Data Analysis from GEO Datasets

Lung Cancer

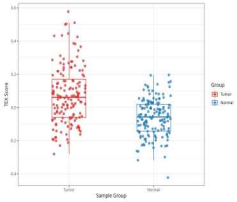
GSE10072
GSE18842
GSE19804
GSE32863



Differential Analysis using limma

Breast Cancer

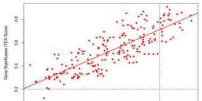
GSE10780
GSE42568
GSE65194
GSE70947



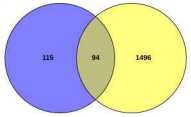
Exhaustion Quantification using ssGSEA

Candidate Gene Identification

Exhaustion scores



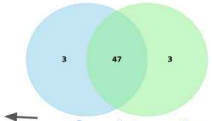
Selection of exhaustion-linked modules using WGCNA



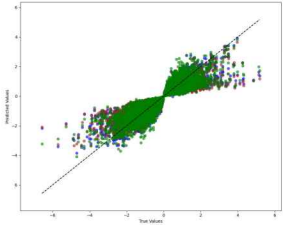
Gene Intersection
WGCNA ∩ DEGs

Conserved Exhaustion Biomarkers

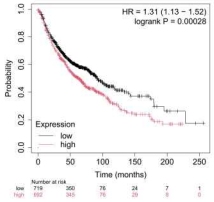
CCL5, HAVCR2, FCGR2A, and FCGR2B regulate immune exhaustion mechanisms.



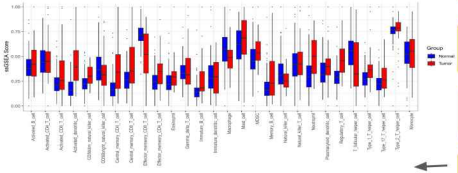
Gene Intersection
RF ∩ LASSO



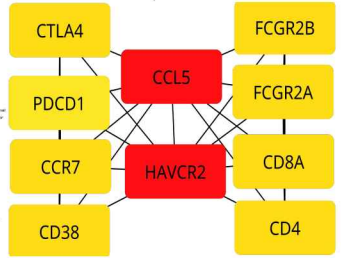
Gene Signature Validation & Identification using Random Forest and LASSO regression



Survival Analysis



Tumor infiltration using ssGSEA & survival analysis



PPI Network analysis using String & Cytoscape

Identification of Hub Genes